SUNHO LEE

https://myshlee417.github.io myshlee417@gmail.com

RESEARCH INTERESTS

I am interested in a secure and efficient architecture of high-performance devices (such as chiplet-based processors, GPUs, NPUs, CXL memories, and PIM devices).

My research objective is to design high-performance devices with security guarantees. To achieve this goal, my recent studies focus on 1) *hardware security* and 2) *performance improvement* of high-performance devices.

Hardware Security of High-performance Devices: As chiplet-based processors, accelerators, CXL memories, and PIM devices are widely used in mission-critical tasks, the importance of security increases. Although I extended Trusted Execution Environment (TEE) to GPUs, NPUs, CXL memories, and heterogeneous processors in prior studies, countless security weaknesses still remain. Therefore, I aim to increase the security level to resist unintended operations of high-performance devices.

Performance Improvement of High-performance Devices: Since machine learning requires speedy processing, I consider both hardware and software optimization to enhance parallelism or to eliminate unnecessary procedures. In my prior studies, I proposed a fine-grained scheduling algorithm in GPUs and NPUs to increase resource utilization. Beyond these optimizations, I target a further improvement as a future research direction.

From these two sub-goals, my objective is to combine a trusted system with a high-performance device design. It is expected to protect users from attackers in a reasonable latency.

PROFESSIONAL SERVICES

University of Oxford, Oxford, United Kingdom *Apr 2026 -*
Oxford OSCAR (Oxford Secure Computer Architecture Research Group) Lab
Oxford HAIGL (Oxford Hardware AI Governance Lab)
Post-doctoral Researcher, Engineering Science
Advisor: Amro Awad

KAIST, Daejeon, Republic of Korea *Mar 2025 - Mar 2026*
KAIST CASYS (KAIST Computer Architecture and SYStem) Lab
Post-doctoral Researcher, School of Computing
Advisor: Jaehyuk Huh

Program Committee: MICRO 2026, ConfAd 2026 (The First Workshop on Advances in Confidential Computing)

Session Chair: ConfAd 2026 (The First Workshop on Advances in Confidential Computing)

External Review Committee: MLSys 2026

Artifact Evaluation Committee: ASPLOS 2026, MICRO 2025, ISCA 2025

Guest Reviewer: TDSC (Transactions on Dependable and Secure Computing), TACO (Transactions on Architecture and Code Optimization), CAL (Computer Architecture Letters)

uArch (in conjunction with **ISCA 2022**), New York City, United States of America *June 2022*
Student Panel
Topic: Life in Grad School

EDUCATION

KAIST, Daejeon, Republic of Korea *Mar 2021 - Feb 2025*
Ph.D. Student, School of Computing
Advisor: Jaehyuk Huh
Thesis: *Trusted Execution Environment for AI Processing System with Heterogeneous Processors and PIM Devices*

KAIST, Daejeon, Republic of Korea *Mar 2019 - Feb 2021*
Master of Science, School of Computing

Advisor: Jaehyuk Huh
Thesis: *Hardware Security Techniques for Trusted Machine Learning Accelerators*

Yonsei University, Seoul, Republic of Korea — *Mar 2015 - Feb 2019*
Bachelor of Science, Computer Science

PUBLICATIONS

- William Taylor, **Sunho Lee**, Henry Wang, Noa Zilberman, and Amro Awad, "VERVE: Verifiable Value Retrieval Leveraging Value Reuse in Accelerator Memory", *the 44th IEEE International Conference on Computer Design (**ICCD**)*, November 2026
- **Sunho Lee**, Minwoo Noh, Jeongwon Choi, Kwanghoon Choi, Igjae Kim, Chang Hyun Park, and Jaehyuk Huh, "SecPIMlet: Confidential Computing for Spatio-temporally Shared Processing-in-Memory", *the 59th IEEE/ACM International Symposium on Microarchitecture (**MICRO**)*, October 2026 **★ Best Paper Nominee**
- **Sunho Lee**, Seonjin Na, Jeongwon Choi, Jinwon Pyo, and Jaehyuk Huh, "Unified Memory Protection with Multi-granular MAC and Integrity Tree for Heterogeneous Processors", *the 52nd International Symposium on Computer Architecture (**ISCA**)*, June 2025
- Kwanghoon Choi, Igjae Kim, **Sunho Lee**, and Jaehyuk Huh, "ShieldCXL: A Practical Obliviousness Support with Sealed CXL Memory", *ACM Transactions on Architecture and Code Optimization (**TACO**)*, March 2025
- Seonjin Na, Jungwoo Kim, **Sunho Lee**, and Jaehyuk Huh, "Supporting Secure Multi-GPU Computing with Dynamic and Batched Metadata Management", *the 30th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*, March 2024
- Jungwoo Kim, Seonjin Na, Sanghyeon Lee, **Sunho Lee**, and Jaehyuk Huh, "Improving Data Reuse in NPU On-chip Memory with Interleaved Gradient Order for DNN Training", *the 56th IEEE/ACM International Symposium on Microarchitecture (**MICRO**)*, October 2023
- *Soojin Hwang, ***Sunho Lee**, Jungwoo Kim, Hongbeen Kim, Jaehyuk Huh, "mNPUsim: Evaluating the Effect of Sharing Resources with Multi-Core NPUs", *the 2023 IEEE International Symposium on Workload Characterization (**IISWC**)*, October 2023 (* co-first authors)
- Seungho Lee, **Sunho Lee**, Jaehyuk Huh, and Sejin Kwon, "Proposal of Aerospace-informatics by Design of Ramjet Inlet Using Machine Learning", *the 2023 Aerospace Europe Conference (**AEC**) joint event between the 10th European Conference for Aerospace Sciences (**EUCASS**) and the 9th Council of European Aerospace Societies (**CEAS**)*, July 2023
- **Sunho Lee**, Seonjin Na, Jungwoo Kim, Jongse Park, and Jaehyuk Huh, "Tunable Memory Protection for Secure Neural Processing Units", *the 40th IEEE International Conference on Computer Design (**ICCD**)*, October 2022
- Seungbeom Choi, **Sunho Lee**, Yeonjae Kim, Jongse Park, Youngjin Kwon, and Jaehyuk Huh, "Serving Heterogeneous Machine Learning Models on Multi-GPU Servers with Spatio-Temporal Sharing", *the 2022 USENIX Annual Technical Conference (**USENIX ATC**)*, July 2022
- **Sunho Lee**, Jungwoo Kim, Seonjin Na, Jongse Park, and Jaehyuk Huh, "TNPU: Supporting Trusted Execution with Tree-less Integrity Protection for Neural Processing Unit", *the 28th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*, April 2022
- Seonjin Na, **Sunho Lee**, Yeonjae Kim, Jongse Park, and Jaehyuk Huh, "Common Counters: Compressed Encryption Counters for Secure GPU Memory", *the 27th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*, March 2021

PATENTS

- **[Application: KR 1020250187986]** Jaehyuk Huh, **Sunho Lee**, Jeongwon Choi, and Jinwon Pyo Jaehyuk Huh, "Device For Protecting Data and Operation Method Thereof", *Korean Patent*
- **[Application: KR 1020250065203]** Jaehyuk Huh, Kwanghoon Choi, Igjae Kim, and **Sunho Lee**, "Efficient Access Obfuscation with CXL Memory", *Korean Patent*

- **[Application: KR 1020230055347]** Jaehyuk Huh, Jungwoo Kim, Seonjin Na, Sanghyeon Lee, and **Sunho Lee**, "Improving the Utilization of NPU On-chip Memory with Computation Rearrangement for DNN Training", *Korean Patent*
- **[Application: KR 1020230055346]** Jaehyuk Huh, Seonjin Na, Jungwoo Kim, and **Sunho Lee**, "Dynamic One-time Pad Table Management for Secure Multi-GPU Communication", *Korean Patent*
- **[Registration: US 12045337]** Jaehyuk Huh, **Sunho Lee**, and Seonjin Na, "Apparatus and Method for Providing Secure Execution Environment for NPU", *US Patent* (with Samsung Electronics)
- **[Application: KR 1020220055977]** Jaehyuk Huh, Seungbeom Choi, Youngjin Kwon, Jongse Park, **Sunho Lee**, and Yeonjae Kim, "Machine Learning Inference Time-spatial SW Scheduler Based on Multiple GPU", *Korean Patent*
- **[Registration: KR 1023652630000]** Jaehyuk Huh, Seonjin Na, **Sunho Lee**, Yeonjae Kim, and Jongse Park, "Efficient Encryption Method and Apparatus for Hardware-based Secure GPU Memory", *Korean Patent*

RESEARCH EXPERIENCES

University of Oxford, Oxford, United Kingdom *Apr 2026 -*
Ongoing Researches at Oxford OSCAR (Oxford Secure Computer Architecture Research Group) Lab
Ongoing Researches at Oxford HAIGL (Oxford Hardware AI Governance Lab)
Advisor: Amro Awad

High-performance Device Security
- Confidential computing for chiplet-based processors
- AI Governance

High-performance Device Performance
- Mitigating the memory wall in GPU via hash caching for value-reuse traffic reduction (Published in **ICCD 2026**)

KAIST, Daejeon, Republic of Korea *Mar 2019 - Mar 2026*
Researches at KAIST CASYS (KAIST Computer Architecture and SYStem) Lab
Advisor: Jaehyuk Huh

High-performance Device Security
- Memory protection optimization for GPU: Common counters for duplicate counters (Published in **HPCA 2021**)
- Inter-processing unit communication protection optimization for multi-GPU: Dynamic OTP table management and batched MAC (Published in **HPCA 2024**)
- Trusted execution environment for NPU: Tensor-granularity counters (Published in **HPCA 2022**)
- Memory protection optimization for NPU: Partial memory protection (Published in **ICCD 2022**)
- Side-channel attack protection for NPU
- Dynamic secure-granularity management for heterogeneous processors: Multi-granular MAC and multi-granular integrity tree (Published in **ISCA 2025**)
- Memory protection and obfuscation for CXL: Sealing CXL module, flit encryption, dummy flit, and DRAM cache partitioning (Published in **TACO 2025**)
- Efficient homomorphic encryption scheme for LLM
- Confidential computing and memory virtualization for multi-tenant PIM (Published in **MICRO 2026**)

High-performance Device Performance
- Multi-tenancy support for a multi-GPU system: Time and spatial sharing (Published in **USENIX ATC 2022**)
- Accurate multi-NPU simulation: Multi-NPU simulator attached with DRAMsim3 (Published in **IISWC 2023**)
- On-chip memory management for training NPU: Access order rearrangement (Published in **MICRO 2023**)

Yonsei University, Seoul, Republic of Korea *Sep 2017 - June 2018*
Undergraduate Research Intern at ELC (Embedded systems Languages and Compilers) Lab
Advisor: Bernd Burgstaller

Parallelism
- Accelerating big-data streaming engine: Multi-thread and shared-memory
- Parallelization of SFA (Simultaneous Deterministic Finite Automata) construction: MPI and Huang's algorithm

RECOGNITION

KAIST, Daejeon, Republic of Korea
Outstanding Teaching Assistant Award - CS311 Computer Organization *Spring 2022, Fall 2019*
Outstanding Teaching Assistant Award - CS230 System Programming *Fall 2023*

Yonsei University, Seoul, Republic of Korea
Dean's List *Spring 2018, Spring 2015*
Undergraduate Capstone Project Award (Third Place) - Project Leader *Spring 2018*
Title: *Cloud SFA: Parallel Construction of Simultaneous Deterministic Finite Automata in Distributed System*

Samsung Electronics, Hwaseong, Republic of Korea
Best Paper Award (Third Place) *Summer 2022*
Title: *TNPU: Supporting Trusted Execution with Tree-less Integrity Protection for Neural Processing Unit*

SKILLS

Programming Languages	C, C++, Python
NPU Simulators	***mNPUsim***, SCALE-Sim, MAESTRO, Gemmini
PIM Simulator/Programming	UPMEM, uPIMulator
GPU Programming	CUDA, MPS
Multi-core CPU Programming	MPI, OpenMP
Machine Learning Frameworks	Pytorch, Tensorflow

TEACHING EXPERIENCES

KAIST, Daejeon, Republic of Korea
Teaching Assistant
CS230 System Programming *Fall 2023, Fall 2021*
CS311 Computer Organization *Fall 2022, Spring 2022, Spring 2021, Fall 2019*
CS211 Digital System and Lab *Spring 2019*
Seminar Presenter
AS602 AI Semiconductor Paper Writing *Spring 2025, Spring 2024*

KAIST Education Center, Daejeon, Republic of Korea
Mentor & Lecturer
Seocho AI College *Summer 2019, Summer 2021*
Python for Beginners *Summer 2022, Summer 2021*